AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science & Technology

Department of Mathematics

MAT2101: Complex Variables, Laplace and Z-transformations (Sections: All)

Final Examination

Total Marks: 40 Time: 2 hours

1.

(a) Find and sketch the path and its orientation is given by $z(t) = 2e^{it}$, $(0 \le t \le \pi)$.

(b) Evaluate $\int_C \text{Im}\{z\}\, dz$ where C is the line segment from $z = 2$ to $z = 2 + i$.

(c) Find the poles and the corresponding residues of $f(z) = \dfrac{9}{z(z-3)}$.

(d) Find $Z\{a^n u[n]\}$, where $u[n]$ is the discrete unit step function.

(e) Find (i) $Z^{-1}\left\{\dfrac{1}{1-2z^{-1}}\right\}, |z| > 2$ and,

(ii) $Z^{-1}\{3z^2\}$

2.

a) Sketch the corresponding path C, where C is the shortest path from 0 to $1 - 2i$ along $z(t) = t - i2t, 0 \le t \le 1$. and hence evaluate $\int_C \text{Re } z\, dz$,.

b) Sketch the path C, where , C is the circle $|z + 2| = 1$, (clockwise) and hence evaluate $\int_C \left(\frac{1}{z+2} - \frac{3}{(z+2)^2}\right) dz$.

3) a) Evaluate $\int_C \dfrac{2z-1}{z(z+2)} dz$, $C: |z + 2| = 3$ using the **Cauchy residue theorem**.

(b)Evaluate $\int_0^{2\pi} \dfrac{d\theta}{5 + 3\sin\theta}$ using the **Cauchy residue theorem.**

(c)Expand $f(z) = \dfrac{3z}{(z-1)(2-z)}$ in a Laurent series valid for $1 < |z| < 2$. Also sketch the ROC.

(d) Evaluate $\int_{-\infty}^{\infty} \dfrac{dx}{x^2 + 4x + 4}$, using contour integration.

4) a) Determine the sequence $x[n]$ by using inverse Z-transform of

$X(z) = (1 + 2z)(1 - 5z^{-1})(1 + 2z^{-2})$. Also sketch $x[n]$.

b) Determine the Z-transform of $(n-2)u[n]$, where $u[n]$ is the discrete time unit step function. Also find and sketch ROC.

c) The difference equation describing the input-output relationship of a discrete-time system with zero initial conditions, is

$$y[n] - 2y[n-1] - 8y[n-2] = x[n] + x[n-1]$$

Compute

(i) the transfer function H(z),

(ii) the discrete-time impulse response $h[n]$.

5)

(a) Use the z-transform to determine $y[n]$, $n \geq 0$ for

$$y[n] - 8y[n-1] + 12y[n-2] = 0, \quad y[-1] = -1,\ y[-2] = 0.$$

(b) Solve by *Z*-transforms, the simultaneous recurrence relation:

$$\boldsymbol{x[n+1] = 2x[n] + 3y[n]},$$
$$y[n+1] = 2x[n] + y[n] \qquad \text{with } x[0] = 1, y[0] = 0.$$